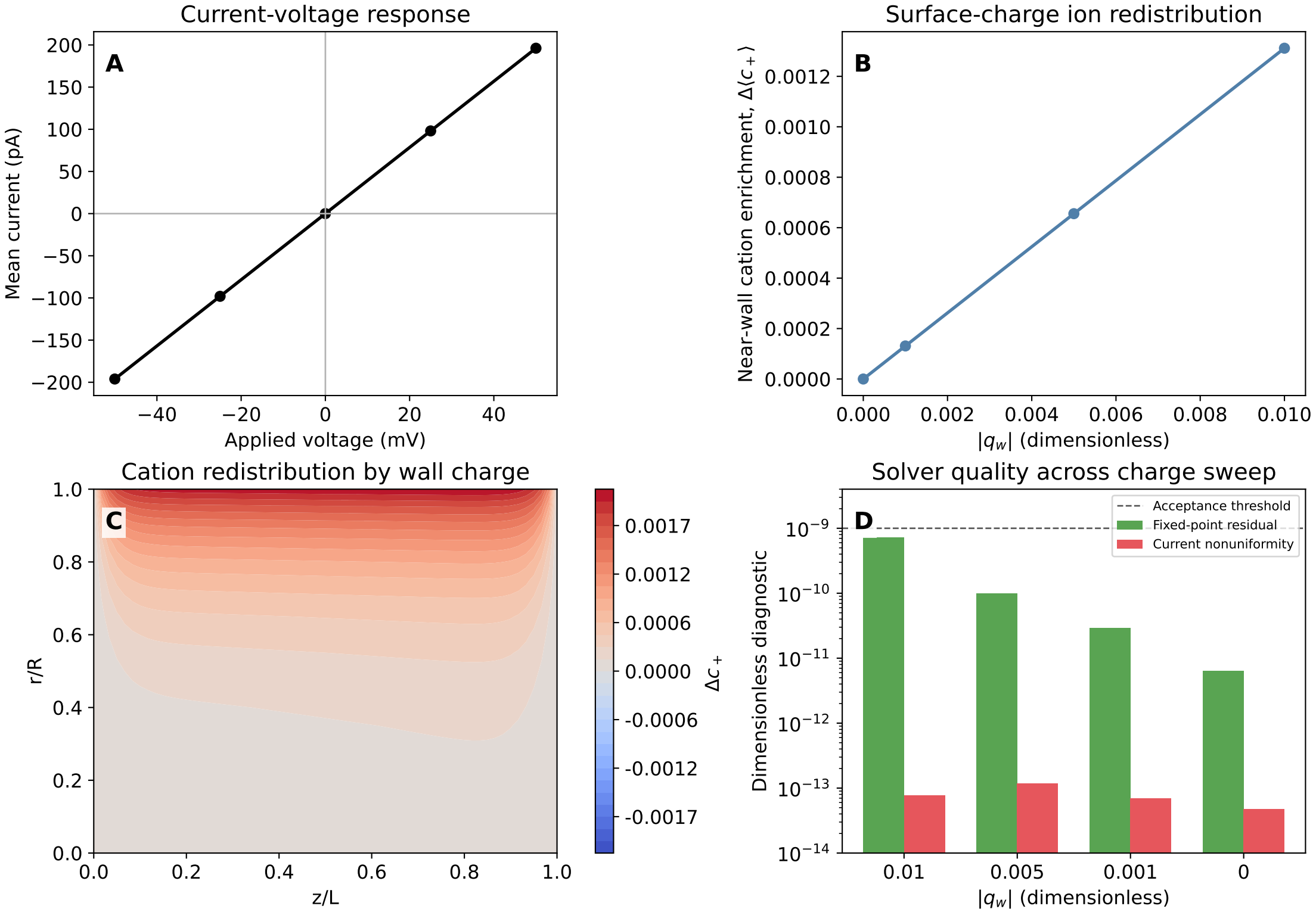
Current-voltage response
A
Mean current (pA)
200
150
100
50
0
−50
−100
−150
−200
−40
−20
0
20
40
Applied voltage (mV)
Surface-charge ion redistribution
B
Near-wall cation enrichment, Δ⟨c+⟩
0.0012
0.0010
0.0008
0.0006
0.0004
0.0002
0.0000
0.000
0.002
0.004
0.006
0.008
0.010
|qw| (dimensionless)
Cation redistribution by wall charge
C
r/R
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
z/L
0.0017
0.0012
0.0006
0.0000
-0.0006
-0.0012
-0.0017
Δc+
Solver quality across charge sweep
D
Acceptance threshold
Fixed-point residual
Current nonuniformity
Dimensionless diagnostic
10−9
10−10
10−11
10−12
10−13
10−14
0.01
0.005
0.001
0
|qw| (dimensionless)